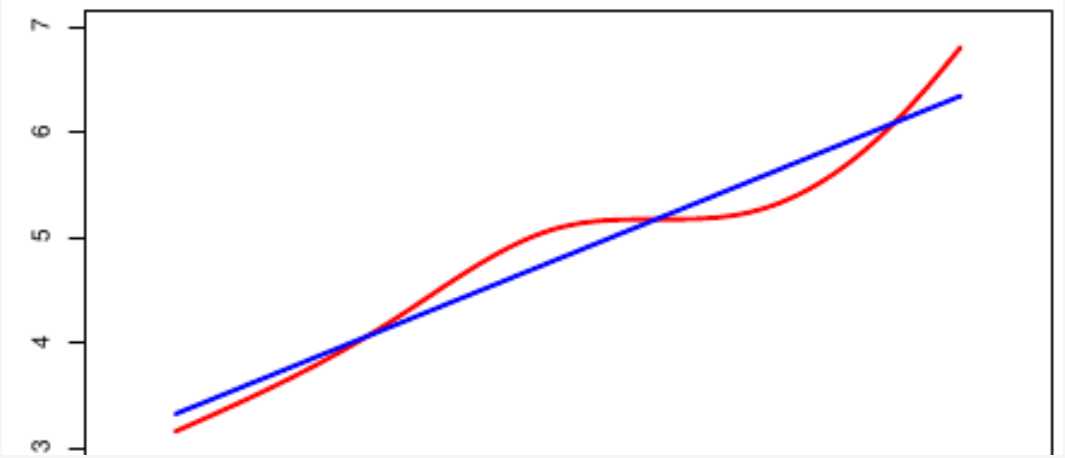
Linear regression

- Linear regression is a simple approach to supervised learning. It assumes that the dependence of Y on $X_1, X_2, \ldots X_p$ is linear.
- True regression functions are rarely exactly linear, but linear models are often useful approximations.



- although it may seem overly simplistic, linear regression is extremely useful both conceptually and practically.

Three different goals

A regression model may be built for at least three different goals:

1. **Prediction:** minimize future prediction error.
2. **Explanation of association:** describe how variables move (change) together, conditional on the chosen model.
3. **Causal inference:** estimate the effect of changing one variable while controlling for confounding and design-related biases.

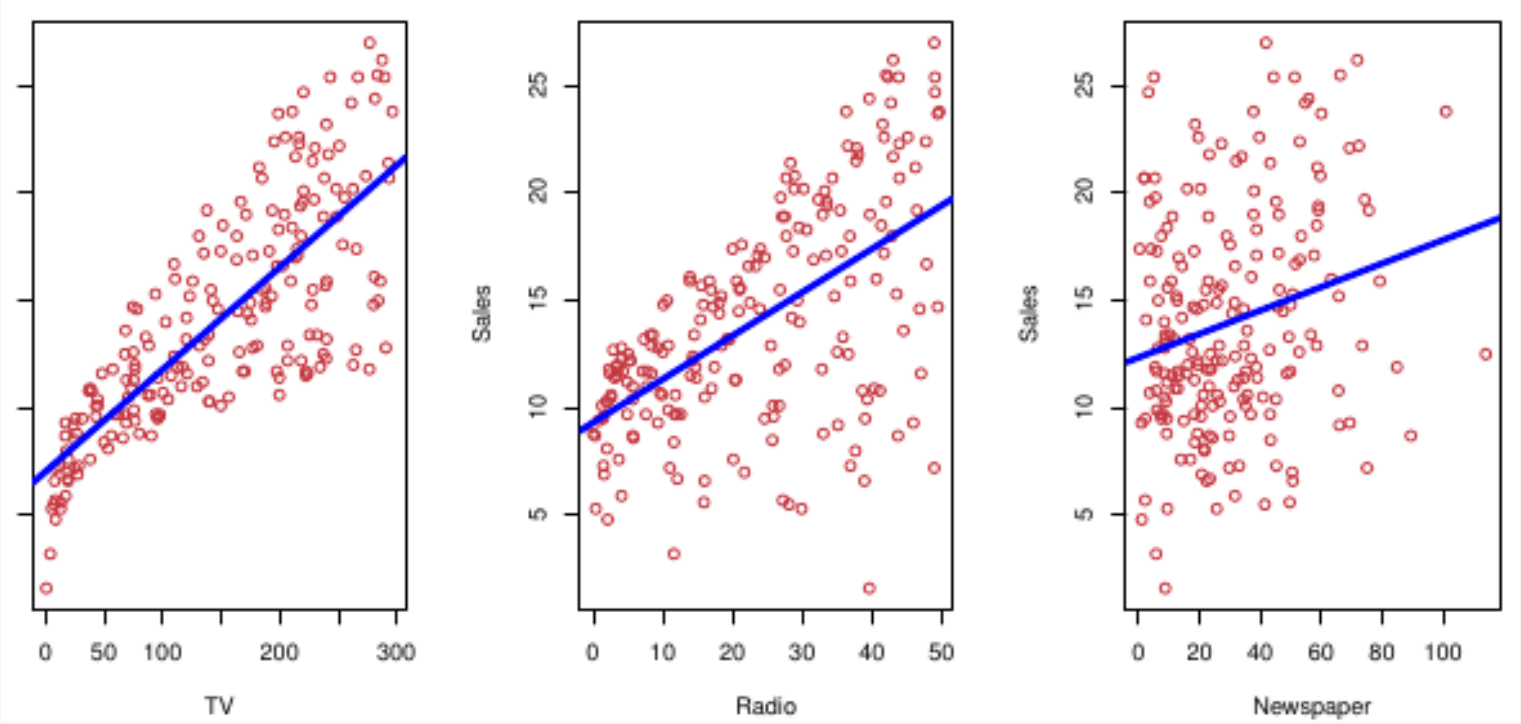
Linear regression for the advertising data

Running example: advertising and sales. Predict product sales from the budgets spent on TV, radio, and newspaper advertising. Consider the advertising data shown on the next slide.

Questions we might ask:

- Is there a relationship between advertising budget and sales?
- How strong is the relationship between advertising budget and sales?
- Which media contribute to sales?
- How accurately can we predict future sales?
- Is the relationship linear?
- Is there synergy among the advertising media?
- Are there better models, are we missing something?

Advertising data



Simple linear regression using a single predictor X.

- We assume a model

$$Y = \beta_0 + \beta_1 X + \epsilon,$$

where β_0 and β_1 are two unknown constants that represent the intercept and slope, also known as coefficients or parameters. ϵ is the error term, for which it holds

$$\mathbb{E}(\varepsilon \mid X) = 0.$$

(For explanation, see the graph of residuals, for a fixed $x = x_0$).

- Given some estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ for the model coefficients, we predict future sales using

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x,$$

where $\hat{y}$ indicates a prediction of Y on the basis of $X = x$. The hat symbol denotes an estimated value.

What are we actually predicting?

A linear regression model does not usually predict the exact observed response for an individual object. Instead, it models the *conditional mean* of the response given the predictors. In simple linear regression, this is written as

$$\mathbb{E}(Y \mid X = x) = \beta_0 + \beta_1 x.$$

For a fixed predictor value x, there may be many possible observed responses Y (see the Figure below). These responses may differ because of noise, measurement error, omitted factors, or natural variation. The regression line represents their average level, not every individual outcome.

What is the meaning of the predicted value?

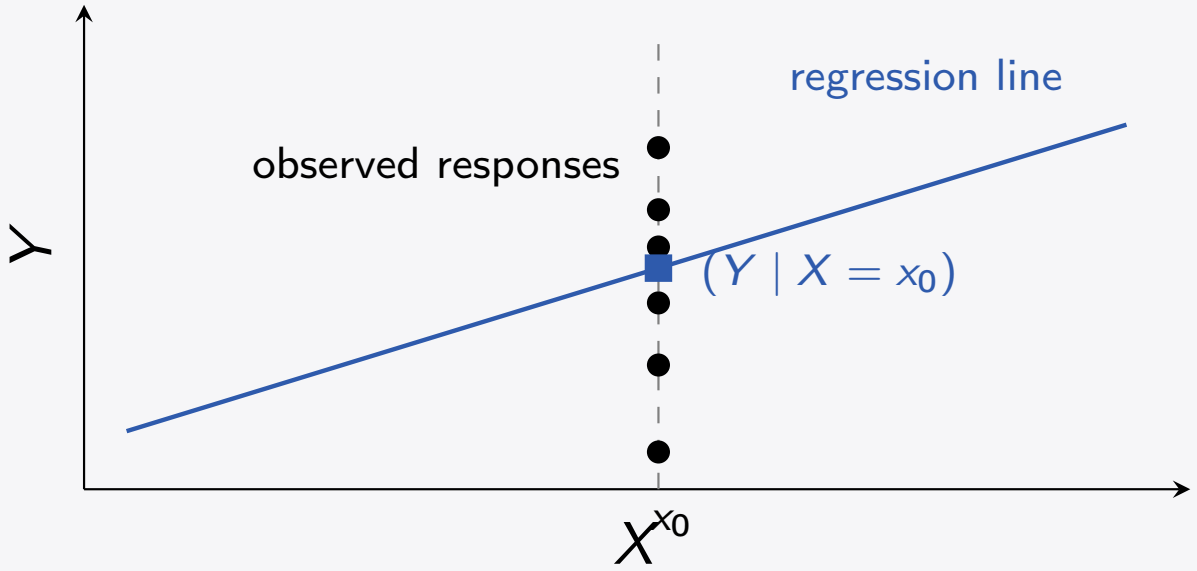


Figure: For a fixed predictor value x_0, several different responses may be observed. Linear regression models their conditional mean $(Y \mid X = x_0)$, not each individual response.

If X is TV budget, then for all campaigns with the same TV budget, actual sales may vary, but the positive and negative deviations from the regression line should average to zero.

Residuals

Let $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ be the prediction for Y based on the ith value of X. Then $e_i = y_i - \hat{y}_i$ represents the ith residual

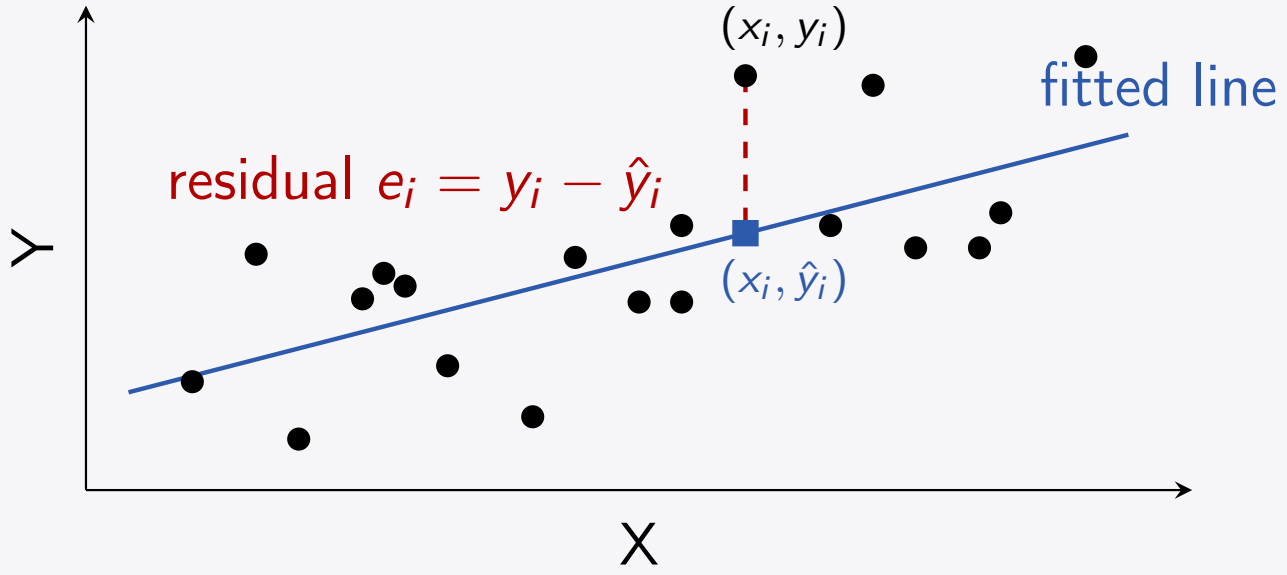


Figure: Simple linear regression. For a given predictor value x_i, the fitted line gives the predicted response $\hat{y}_i$. The residual is the difference between what we observed and what the model predicted, that is, $e_i = y_i - \hat{y}_i$.

Estimation of the parameters by least squares

- We define the residual sum of squares (RSS) as

$$\text{RSS} = e_1^2 + e_2^2 + \cdots + e_n^2,$$

or equivalently as

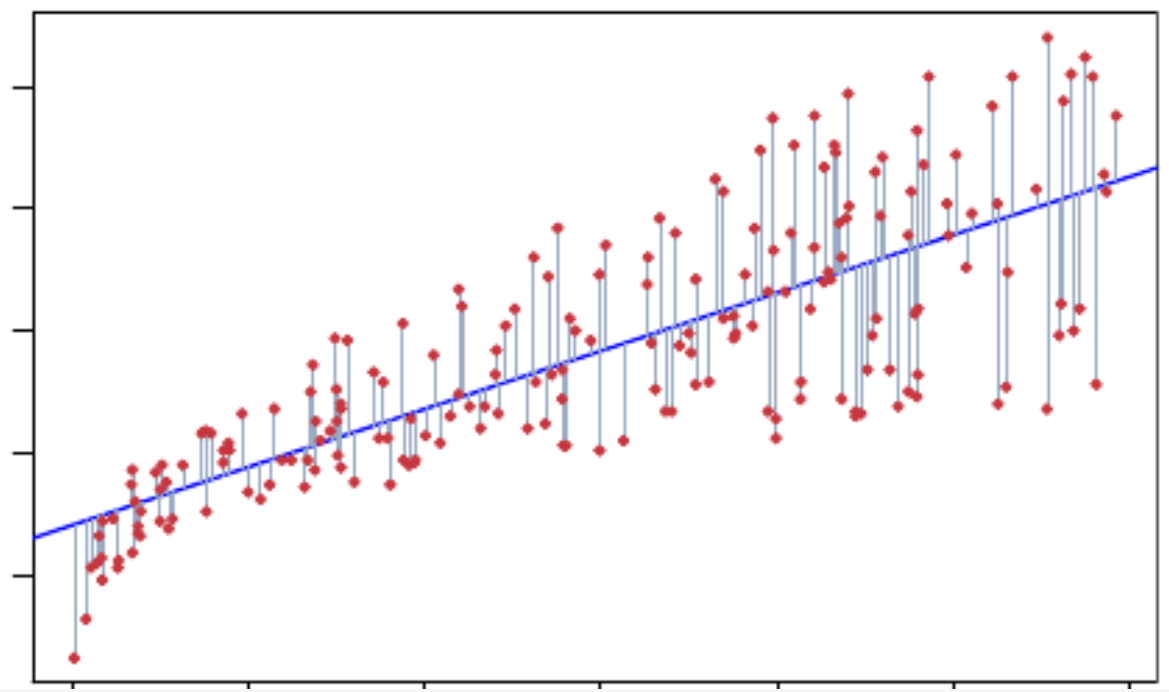
$$\text{RSS} = (y_1 - \hat{\beta}_0 - \hat{\beta}_1 x_1)^2 + (y_2 - \hat{\beta}_0 - \hat{\beta}_1 x_2)^2 + \ldots + (y_n - \hat{\beta}_0 - \hat{\beta}_1 x_n)^2.$$

- The least squares approach chooses $\hat{\beta}_0$ and $\hat{\beta}_1$ to minimize the RSS. The minimizing values can be shown to be

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}, \tag{1}$$

where $\bar{y} \equiv \frac{1}{n}\sum_{i=1}^n y_i$ and $\bar{x} \equiv \frac{1}{n}\sum_{i=1}^n x_i$ are the sample means. These equations give least-square estimates for the coefficients, from the corresponding estimators.

Example: advertising data



The least squares fit for the regression of `sales` onto `TV`.
In this case a linear fit captures the essence of the relationship, although it is somewhat deficient in the left of the plot.

Assessing the accuracy of the coefficient estimates

- The standard error of an estimator reflects how it varies under repeated sampling. We have

$$\mathrm{SE}(\hat{\beta}_1)^2 = \frac{\sigma^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}, \qquad \mathrm{SE}(\hat{\beta}_0)^2 = \sigma^2 \left[\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}\right],$$

where $\sigma^2 = \mathrm{Var}(\epsilon)$

- These standard errors can be used to compute confidence intervals. A 95% confidence interval is produced by a method that, over repeated sampling, contains the true parameter in 95% of samples. It has the form

$$\hat{\beta}_1 \pm 2 \cdot \mathrm{SE}(\hat{\beta}_1).$$

Confidence intervals — continued

The confidence interval

$$\left[\hat{\beta}_1 - 2 \cdot \mathrm{SE}(\hat{\beta}_1),\ \hat{\beta}_1 + 2 \cdot \mathrm{SE}(\hat{\beta}_1)\right]$$

will contain the true value of β_1 in 95% of the sampled confidence intervals.

For the advertising data, the 95% confidence interval for β_1 is

$$[0.042, 0.053]$$

This confidence interval tells us how uncertain we are about the slope coefficient.

Confidence vs. prediction interval

We can also calculate the confidence interval about the average response, and the prediction interval about an individual future outcome.

- The confidence interval gives a plausible range for the mean response at a chosen predictor value. It is used when we want to estimate the average outcome, that is, the height of the regression line at that point. In the sales - TV example, it tells us where the average sales level is likely to be for campaigns with a given TV budget. So it helps us understand the underlying trend in the data: does spending more on TV tend to move average sales up, and by how much?
- The prediction interval gives a plausible range for one new individual observation at that predictor value. It is used when we want to predict what may happen for a single future case.
  It tells us how far an individual campaign's sales might fall above or below that average. So it helps us understand the practical uncertainty of prediction: even if the average sales at this budget are, say, 100, one actual campaign may still end up at 85 or 115.

Confidence vs prediction interval

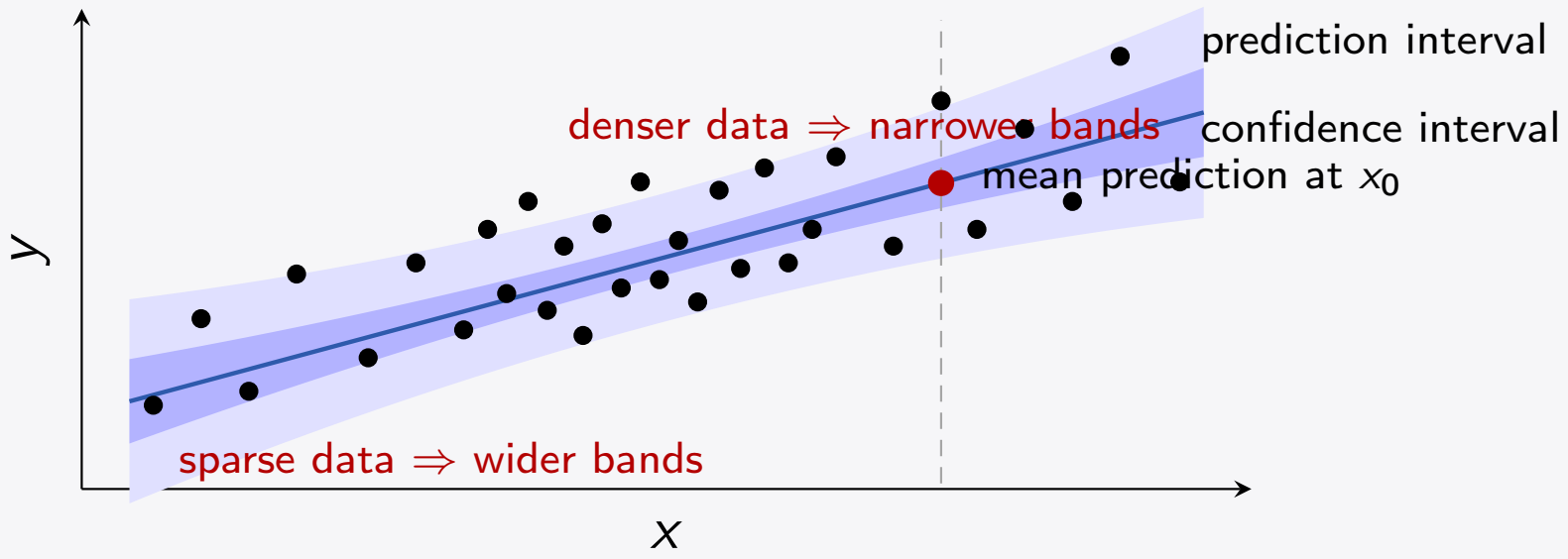


Figure: The confidence interval for the mean response is narrowest where the data are most concentrated and wider where the predictor values are sparse. The prediction interval is wider because it targets a new individual observation. Individual points may fall outside the confidence interval, and some may even fall outside the prediction interval.

Important. Predictions outside the observed predictor range are extrapolations and are much less reliable.

Hypothesis testing

- Standard errors can also be used to perform hypothesis tests on the coefficients. The most common hypothesis test involves testing the null hypothesis of

$$H_0 : \text{ There is no relationship between } X \text{ and } Y$$

versus the alternative hypothesis

$$H_A : \text{ There is some relationship between } X \text{ and } Y.$$

- Mathematically, this corresponds to testing

$$H_0 : \beta_1 = 0 \qquad \text{versus} \qquad H_A : \beta_1 \neq 0,$$

since if $\beta_1 = 0$ then the model reduces to $Y = \beta_0 + \epsilon$, and X is not associated with Y.

Hypothesis testing - continued

- To test the null hypothesis, we compute a t-statistic, given by

$$t = \frac{\hat{\beta}_1 - 0}{\mathrm{SE}(\hat{\beta}_1)},$$

- This will have a t-distribution with $n - 2$ degrees of freedom, assuming $\beta_1 = 0$.
- Using statistical software, it is easy to compute the probability of observing any value equal to $|t|$ or larger. We call this probability the p-value.
- In multiple linear regression, individual coefficients are still often tested with t-tests, while the overall model can be tested with an F-test (ANOVA type of a test). For the simple linear regression it holds $F = t^2$.

Results for the advertising data

	Coefficient	Std. Error	t-statistic	p-value
Intercept	7.0325	0.4578	15.36	< 0.0001
TV	0.0475	0.0027	17.67	< 0.0001

- The estimated slope for TV is positive and highly statistically significant, indicating that larger TV budgets are associated with higher average sales. In particular, a one-unit increase in TV budget is associated with an average increase of 0.0475 units in sales. Be careful with units: if TV is measured in thousands of dollars, then a one-unit increase means an additional 1000 dollars.
- If TV=0, then predicted sales are about 7.03. This means that even with no TV advertising, the model still expects some baseline level of sales, which may come from other reasons (brand recognition, general market demand etc.) In some models the predictor can not take the value 0 (it is outside of the predictor's range, so such an interpretation would not be meaningful).

Assessing the overall accuracy of the model

- We compute the Residual Standard Error

$$\mathrm{RSE} = \sqrt{\frac{1}{n-2}\mathrm{RSS}} = \sqrt{\frac{1}{n-2}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2},$$

where the residual sum-of-squares is $\mathrm{RSS} = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$.

- R-squared or fraction of variance explained is

$$R^2 = \frac{\mathrm{TSS} - \mathrm{RSS}}{\mathrm{TSS}} = 1 - \frac{\mathrm{RSS}}{\mathrm{TSS}}$$

where $\mathrm{TSS} = \sum_{i=1}^{n}(y_i - \bar{y})^2$ is the total sum of squares.

- It can be shown that in this simple linear regression setting that $R^2 = r^2$, where r is the correlation between X and Y:

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}.$$

Advertising data results

Quantity	Value
Residual Standard Error	3.26
R^2	0.612
F-statistic	312.1

The fitted regression gives, for each TV budget, a predicted average sales value on the regression line. The residual standard error tells us how far the actual observed sales typically lie from those predicted values. Here, this typical prediction error is about 3.26 sales units. Whether 3.26 is large or small depends on the scale of the sales values.

Advertising data results

- Some of the variation in Y can be explained by variation in the X's and some cannot.
- R^2 tells you the fraction of variance that can be explained by X.
- R^2 is always between 0 and 1. Zero means no variance has been explained. One means it has all been explained (perfect fit to the data).
- However, this measure increases when we add new predictors even if they are not significant or irrelevant, so it is should be not used alone.
- In our example $R^2 = 0.612$ This means that about 61.2% of the variability in sales is explained by the linear relationship with TV budget in this model. The remaining variation is due to other factors or random variability not captured by the model.

Interpretation of R^2

For a simple linear model $R^2 = r^2$, so calculating the correlation can tell us a lot about the linear model. For example, if $r = 0.2$ then $R^2 = 0.04$ - such a model will explain only 4% of the variability. The coefficient β_1 can still be significant, it may still correspond to a weak relationship, especially in large samples. We can continue by checking effect size and do residual diagnostics.
Something similar may happen in multidimensional models, when we should check for multicollinearity, confounding etc...

Multiple Linear Regression

- Multiple linear regression algorithms are used to predict the response variable Y, for a set of explanatory variables $X_1, X_2, ..., X_p$. The model is

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \epsilon,$$

where ε is an error term (random variable) satisfying

$$\mathbb{E}(\varepsilon \mid X_1, \ldots, X_p) = 0.$$

- β_0 is the intercept (i.e. the average value for Y if all the X's are zero, in case those are possible values for the variables).
- For $j \neq 0$, we interpret β_j as the average effect on Y of a one unit increase in X_j, holding all other predictors fixed. In the advertising example, the model becomes

$$\texttt{sales} = \beta_0 + \beta_1 \times \texttt{TV} + \beta_2 \times \texttt{radio} + \beta_3 \times \texttt{newspaper} + \epsilon.$$

Interpreting regression coefficients

Interpretation is easiest when predictors are weakly correlated or when the design allows their effects to be separated clearly:

- Each coefficient can be estimated and tested separately.
- Interpretations such as "a unit change in X_j is associated with a β_j change in Y, while all the other variables stay fixed", are possible.

Correlations amongst predictors cause problems:

- The variance of all coefficients tends to increase, sometimes dramatically
- Interpretations become hazardous - when X_j changes, everything else changes.

Claims of causality should be avoided for observational data.

Interpreting regression coefficients

Example. Suppose we have a fruit basket, with bananas and mangoes.
Let Y be total price of a fruit basket,
X_1 be total number of fruits in the basket,
X_2 be number of bananas in the basket.
If we regress Y only on X_2, then the coefficient of X_2 describes the overall association between the number of bananas and the basket price. In simple regression, the sign of the coefficient of X_2 depends on the overall pattern in the data. In multiple regression, after holding X_1 fixed, the interpretation becomes clearer.
If we regress Y on both X_1 and X_2, then the coefficient of X_2 is interpreted *while holding the total number of fruits fixed*. In that case, increasing the number of bananas means decreasing the number of mangoes which are more expensive. Thus, Y would decrease, and the coefficient of X_2 will be negative.

Estimation and Prediction for Multiple Regression

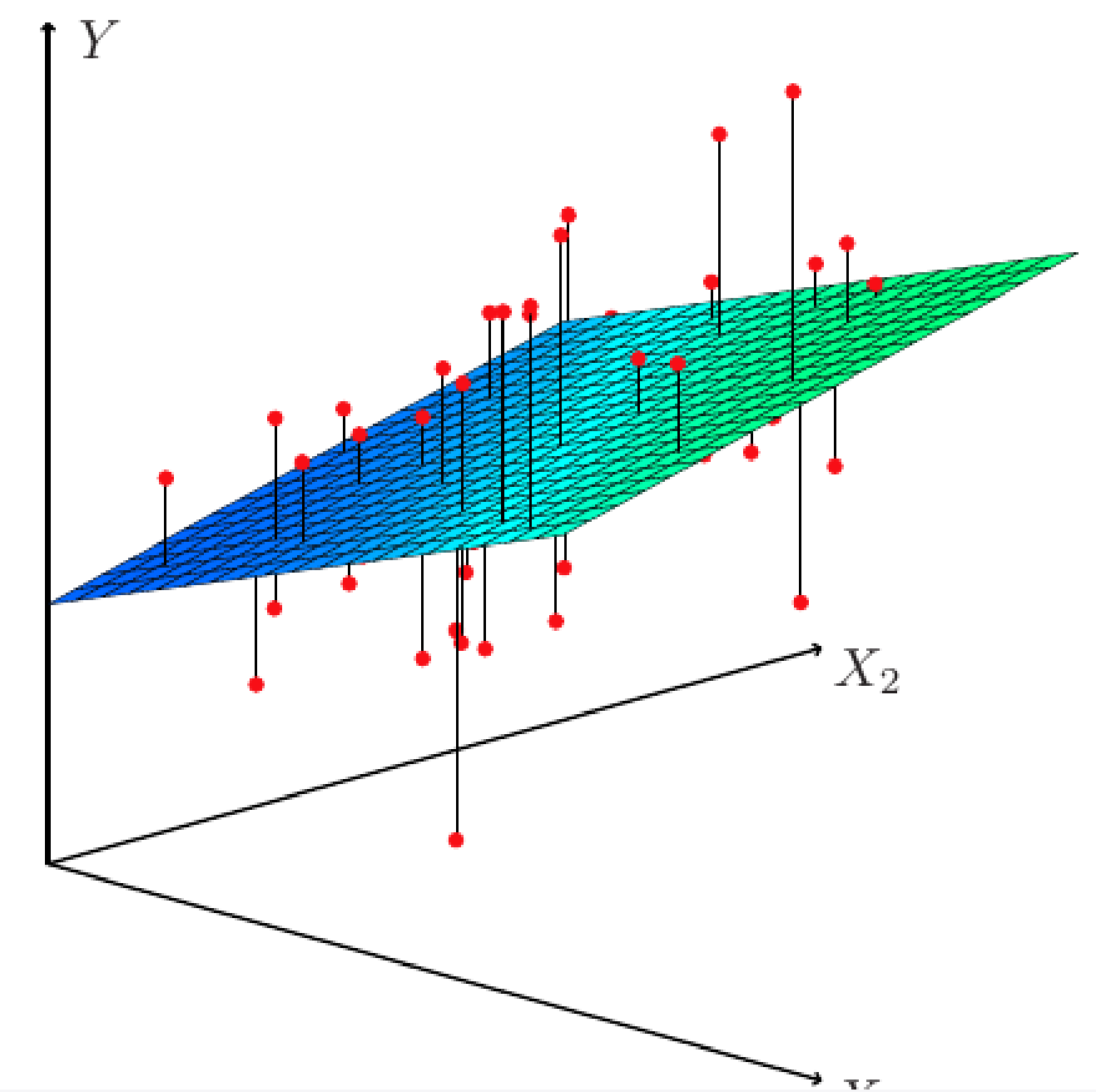
- Given estimates $\hat{\beta}_0, \hat{\beta}_1, \ldots \hat{\beta}_p$, we can make predictions using the formula

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \cdots + \hat{\beta}_p x_p.$$

- We estimate $\beta_0, \beta_1, \ldots, \beta_p$ as the values that minimize the sum of squared residuals

$$\text{RSS} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \hat{\beta}_2 x_{i2} - \cdots - \hat{\beta}_p x_{ip})^2.$$

This is done using standard statistical software. The values $\hat{\beta}_0, \hat{\beta}_1, \ldots, \hat{\beta}_p$ that minimize RSS are the multiple least squares regression coefficient estimates.


Y
X_2

Some important questions

1. Is $\beta_j = 0$ or not? We can use a hypothesis test to answer this question. If we can't be sure that $\beta_j \neq 0$, should we use X_j as one of our predictors? (yes it may still be useful for predicting, to control confounding etc.)

2. Can we be sure that at least one of our X variables is a useful predictor i.e. is it the case that

$$\beta_1 = \beta_2 = \cdots = \beta_p = 0?$$

3. Do all the predictors help to explain Y, or is only a subset of the predictors useful?
4. How well does the model fit the data?
5. Given a set of predictor values, what response value should we predict, and how accurate is our prediction?

1. Is $\beta_j = 0$ i.e. is X_j an important variable?

- We use a hypothesis test to answer this question $H_0: \ \beta_j = 0$ vs $H_a: \ \beta_j \neq 0$
- Calculate

$$t = \frac{\hat{\beta}_j}{SE(\hat{\beta}_j)}$$

  Number of standard deviations away from zero.

- If t is large (equivalently p-value is small) we have evidence that $\beta_j \neq 0$, i.e. the data X_j is associated with Y, under the model assumptions.

Results for advertising data

	Coefficient	Std. Err	t-value	p-value
Constant	12.3514	0.6214	19.8761	0.0000
Newspaper	0.0547	0.0166	3.2996	0.0011

Table: Regression coefficients

The slope term coefficient here means that for a \$1000 increase in spending on newspaper advertising is associated with an average increase in sales by around 55 units (in the given data the predictor variables are in thousands of dollars and sales are in thousands of units)

	Coefficient	Std. Error	t-statistic	p-value
Intercept	2.939	0.3119	9.42	< 0.0001
TV	0.046	0.0014	32.81	< 0.0001
radio	0.189	0.0086	21.89	< 0.0001
newspaper	-0.001	0.0059	-0.18	0.8599

For a given amount of TV and Newspaper advertising (holding fixed), spending an additional \$1000 on radio advertising leads to increase in sales by approximately 189 units.

Comparing the results

Why did the significance of the Newspaper variable change (the p value was small in the simple regression and it was not significant in the multiple regression) – In the simple regression we ignore the other predictors (in this case TV and radio) and in the multiple regression they are included and hold fixed, but still they explain a lot of the variability.

Let us inspect on the data more, check the correlation coefficients:

	TV	radio	newspaper	sales
TV	1.0000	0.0548	0.0567	0.7822
radio		1.0000	0.3541	0.5762
newspaper			1.0000	0.2283
sales				1.0000

we will see that there is a tendency to spend more in newspaper advertising in markets where it is spent more on radio. Thus "newspaper" gets "credit" for the effect of radio predictor on sales. Thus if the the multiple regression is correct, after adjusting for TV and radio, newspaper does not show an additional linear association with sales in this model. The variability is explained by the other two variables.

Is the whole regression explaining anything at all - deciding for important variables

Check the p-value of the F-test. If it is significant, then based on our data we can reject H_0 in favor of the alternative hypothesis H_a. What next?
Test for:

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_p = 0, \qquad H_a : \text{at least one slope } \neq 0.$$

Source	df	SS	MS	F	p-value
Explained	2	4860.2347	2430.1174	859.6177	0.0000
Unexplained	197	556.9140	2.8270		

The answer comes from the F-test in the ANOVA (**AN**alysis **O**f **VA**riance) table.
The ANOVA table contains several quantities, but we are interested in the value of the F-**statistic** and its corresponding p-**value**.

Deciding on important variables - variable selection

One important step is to examine whether the overall regression explains variation in the response, for example using the F-test. If we reject the H_0 hypothesis, this means that at least one of the predictors is related to the response, and next we should decide which ones.

Variable selection is very important and there are different approaches to choose the variables for the model (forward, backward etc.) Often we have to check for multicollinearity and reduce the dimensionality of data.

Model quality can be assessed using criteria such as Akaike information, Bayesian information criterion, adjusted R^2, Cross-validation.

Deciding on the important variables

- The most direct approach is called *all subsets* or *best subsets* regression: we compute the least squares fit for all possible subsets and then choose between them based on some criterion that balances training error with model size.
- However we often can't examine all possible models, since they are 2^p of them; for example when $p = 40$ there are over a billion models!

Instead we need an automated approach that searches through a subset of them. We discuss two commonly used approaches next.

Forward selection

- Begin with the null model - a model that contains an intercept but no predictors.
- Fit p simple linear regressions and add to the null model the variable that results in the lowest RSS.
- Add to that model the variable that results in the lowest RSS amongst all two-variable models.
- Continue until some stopping rule is satisfied. The stopping rule should preferably be based on validation, adjusted R^2, AIC/BIC, or cross-validation rather than only p-values."

Backward selection

- Start with all variables in the model.
- Remove the variable with the largest p-value - that is, the variable that is the least statistically significant.
- The new $(p-1)$-variable model is fit, and the variable with the largest p-value is removed.
- Continue until a stopping rule is reached. For instance, we may stop when all remaining variables have a significant p-value defined by some significance threshold.

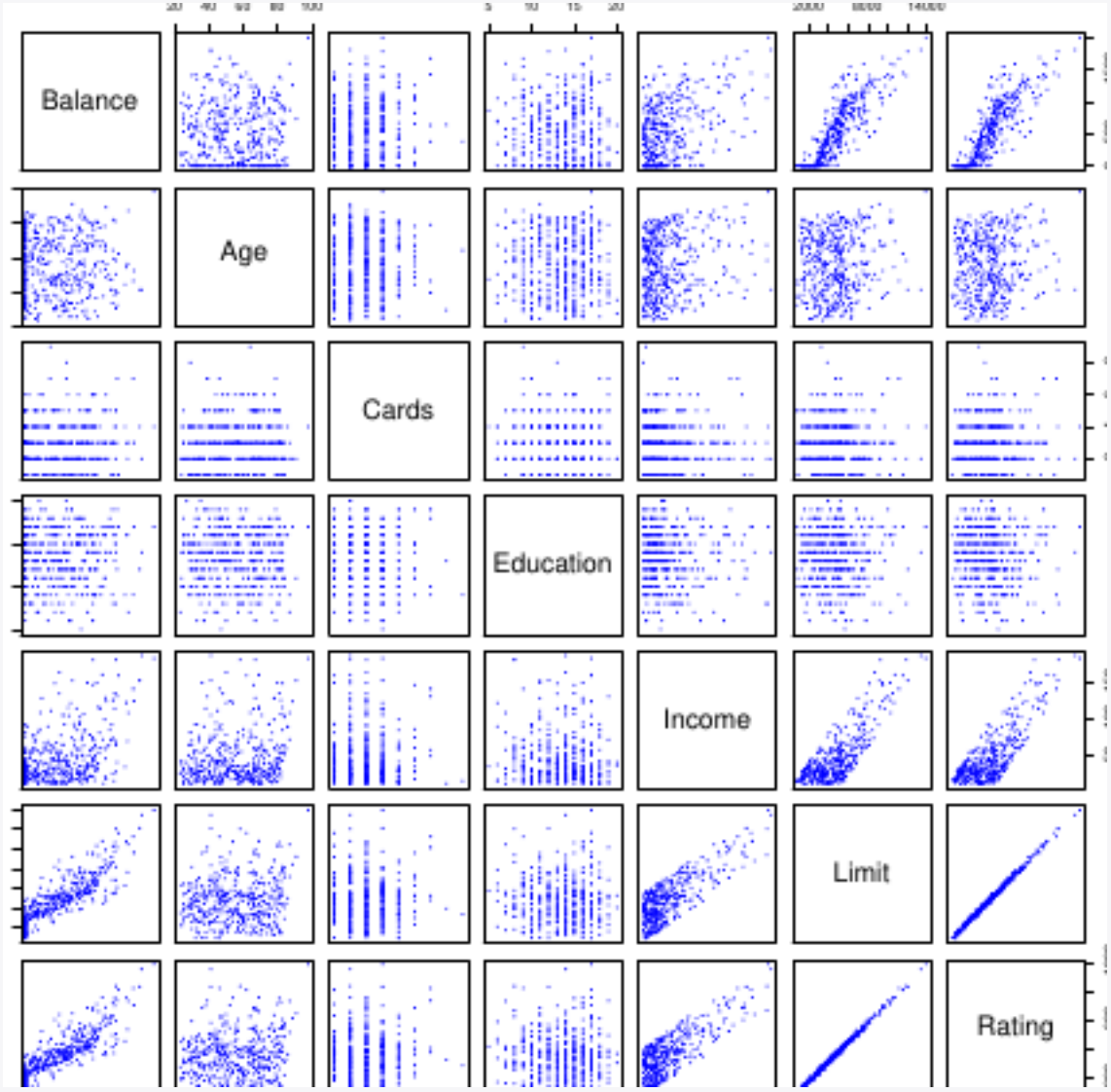
Other Considerations in the Regression Model

Qualitative Predictors

- Some predictors are not quantitative but are qualitative, taking a discrete set of values.
- These are also called categorical predictors or factor variables.
- See for example the scatterplot matrix of the credit card data in the next slide.

In addition to the 7 quantitative variables shown, there are four qualitative variables: `gender`, `student` (student status), `status` (marital status), and `ethnicity` (Caucasian, African American (AA) or Asian).

Credit Card Data



Interpretation

- Suppose we want to include income and gender (male and female). Let

$$\text{Gender}_i = \begin{cases} 0 & \text{if male,} \\ 1 & \text{if female.} \end{cases}$$

- Then the regression equation is

$$Y_i \approx \beta_0 + \beta_1 \text{Income}_i + \beta_2 \text{Gender}_i = \begin{cases} \beta_0 + \beta_1 \text{Income}_i & \text{if male,} \\ \beta_0 + \beta_1 \text{Income}_i + \beta_2 & \text{if female.} \end{cases}$$

  This creates 2 parallel regression lines (same slope) - see the figures in the slides below.

- β_2 is the average difference in monthly balance between females and males, for a given income level. Males are the "baseline".

Variable	Coefficient	Std. Err	t-value	p-value
Constant	233.7663	39.5322	5.9133	0.0000
Income	0.0061	0.0006	10.4372	0.0000
Gender_Female	24.3108	40.8470	0.5952	0.5521

Other Coding Schemes

- There are different ways to code categorical variables.
- Two genders (male and female). Let

$$\boxed{Gender_i = \begin{cases} -1 & \text{if male} \\ 1 & \text{if female} \end{cases}}$$

then the regression equation is

$$\boxed{Y_i \approx \beta_0 + \beta_1 Income_i + \beta_2 Gender_i = \begin{cases} \beta_0 + \beta_1 Income_i - \beta_2, & \text{if male} \\ \beta_0 + \beta_1 Income_i + \beta_2, & \text{if female} \end{cases}}$$

- Under this coding, β_0 represents the grand mean at a given income level, and $2\beta^2$ is the difference between the two groups.β_2 is the average amount that females are above the average, for any given income level. β_2 is also the average amount that males are below the average, for any given income level (assuming β_2 is positive).

Qualitative predictors with more than two levels

- With more than two levels, we create additional dummy variables. For example, for the `ethnicity` variable we create two dummy variables. The first could be

$$x_{i1} = \begin{cases} 1 & \text{if the } i\text{th person is Asian} \\ 0 & \text{if the } i\text{th person is not Asian,} \end{cases}$$

and the second could be

$$x_{i2} = \begin{cases} 1 & \text{if the } i\text{th person is Caucasian} \\ 0 & \text{if the } i\text{th person is not Caucasian.} \end{cases}$$

Qualitative predictors with more than two levels — continued.

- Then both of these variables can be used in the regression equation, in order to obtain the model

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \epsilon_i = \begin{cases} \beta_0 + \beta_1 + \epsilon_i & \text{if the } i\text{th person is Asian} \\ \beta_0 + \beta_2 + \epsilon_i & \text{if the } i\text{th person is Caucasian} \\ \beta_0 + \epsilon_i & \text{if the } i\text{th person is AA.} \end{cases}$$

- There will always be one fewer dummy variable than the number of levels. The level with no dummy variable - African American in this example - is known as the baseline.

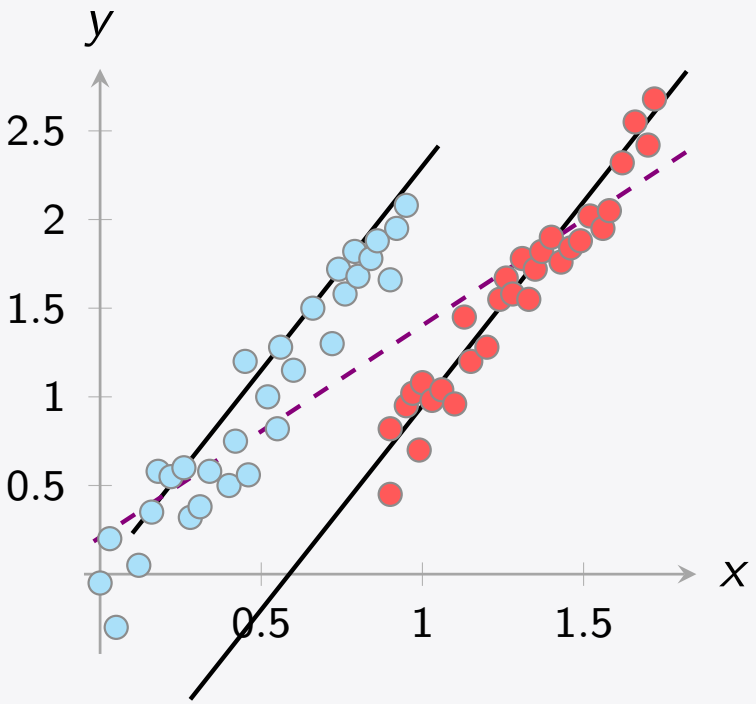
Results for ethnicity

Coefficient	Std. Error	t-statistic	p-value	
Intercept	531.00	46.32	11.464	< 0.0001
ethnicity[Asian]	-18.69	65.02	-0.287	0.7740
ethnicity[Caucasian]	-12.50	56.68	-0.221	0.8260

Neither ethnicity coefficient is statistically significant relative to the baseline group, so this model does not provide evidence of a difference in mean balance by ethnicity after coding with AA as the baseline.

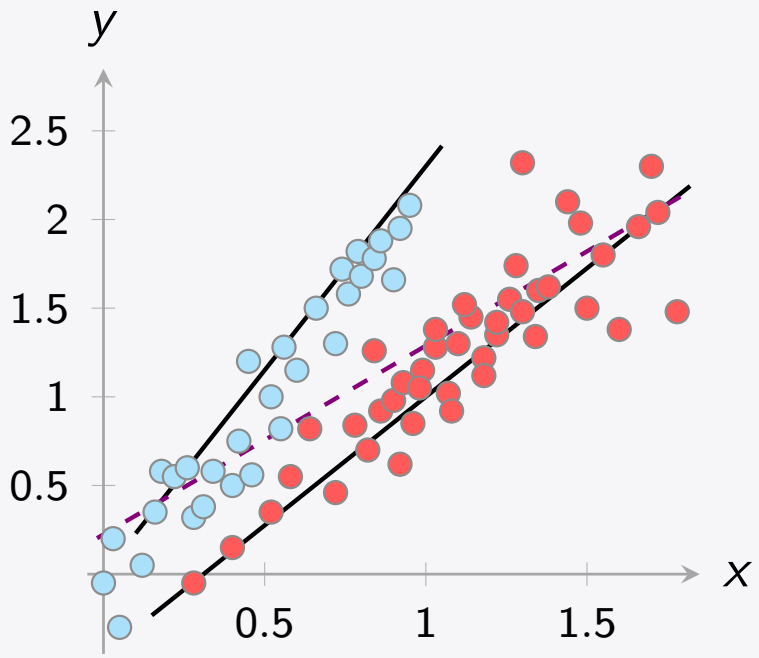
Including interaction effects in the model

What is the difference and what should we include in the second model?



same slopes

vs.



different slopes

Interactions and nonlinearity

- Example. In our previous analysis of the Advertising data, we assumed that the effect on sales of increasing one advertising medium is independent of the amount spent on the other media. For example, the linear model

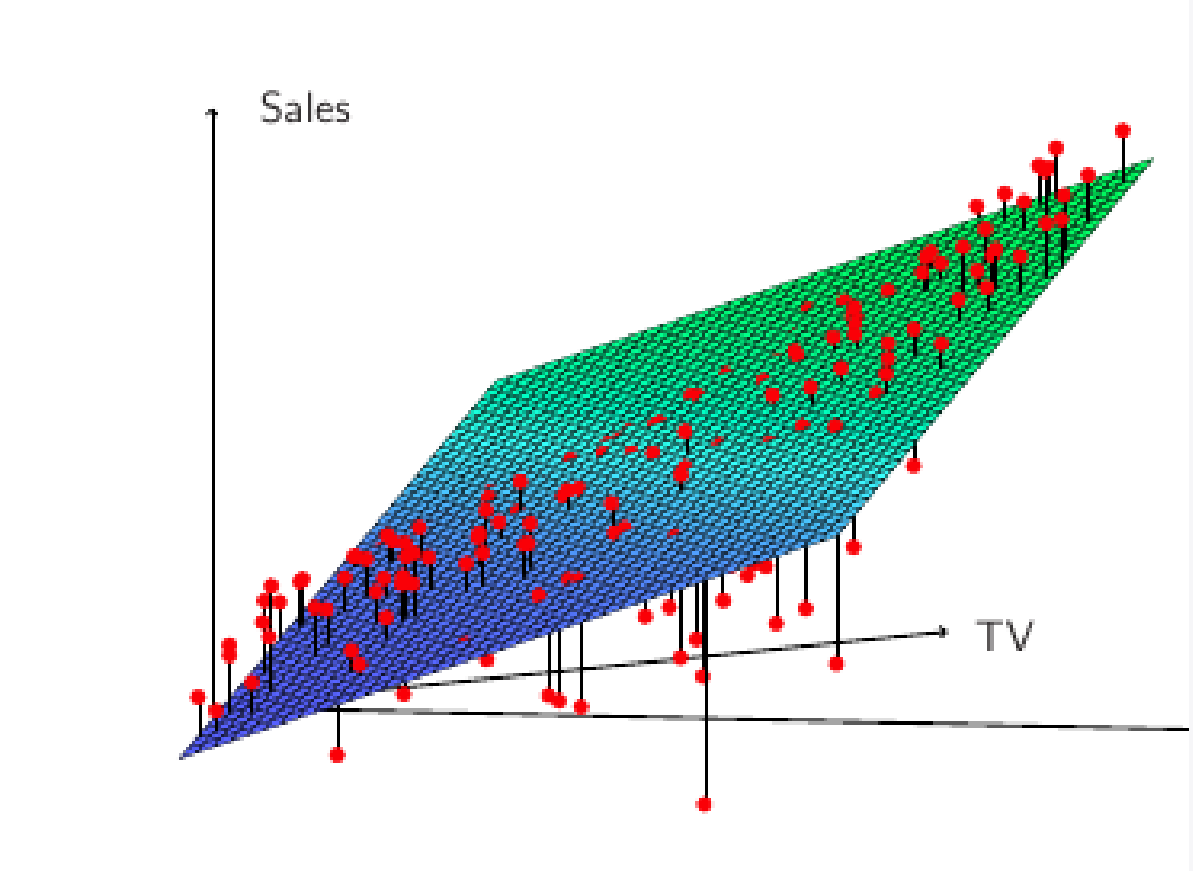
$$\texttt{sales} = \beta_0 + \beta_1 \times \texttt{TV} + \beta_2 \times \texttt{radio} + \beta_3 \times \texttt{newspaper}$$

  states that the average effect on sales of a one-unit increase in TV is always β_1, regardless of the amount spent on radio.

Interactions — continued

- But suppose that spending money on radio advertising actually increases the effectiveness of TV advertising, so that the slope term for TV should increase as radio increases.
- In this situation, given a fixed budget of \$100,000, spending half on radio and half on TV may increase sales more than allocating the entire amount to either TV or to radio.
- In marketing, this is known as a synergy effect, and in statistics it is referred to as an interaction effect.

Interaction in the Advertising data?



When levels of either `TV` or `radio` are low, then the true `sales` are lower than predicted by the linear model.
But when advertising is split between the two media, then the model tends to underestimate `sales`.

Modelling interactions — Advertising data

Model takes the form

$$\texttt{sales} = \beta_0 + \beta_1 \times \texttt{TV} + \beta_2 \times \texttt{radio} + \beta_3 \times (\texttt{radio} \times \texttt{TV}) + \epsilon$$

$$= \beta_0 + (\beta_1 + \beta_3 \times \texttt{radio}) \times \texttt{TV} + \beta_2 \times \texttt{radio} + \epsilon.$$

Results:

Coefficient	Std. Error	*t*-statistic	*p*-value	
Intercept	6.7502	0.248	27.23	< 0.0001
TV	0.0191	0.002	12.70	< 0.0001
radio	0.0289	0.009	3.24	0.0014
TV×radio	0.0011	0.000	20.73	< 0.0001

Interpretation

- The results in this table suggests that interactions are important.
- The p-value for the interaction term `TV×radio` is extremely low, indicating that there is strong evidence for $H_A : \beta_3 \neq 0$.
- The R^2 for the interaction model is 96.8%, compared to only 89.7% for the model that predicts sales using TV and radio without an interaction term.

Interpretation — continued

- This means that $(96.8 - 89.7)/(100 - 89.7) = 69\%$ of the variability in sales that remains after fitting the additive model has been explained by the interaction term.
- The coefficient estimates in the table suggest that an increase in TV advertising of \$1,000 is associated with increased sales of

$$(\hat{\beta}_1 + \hat{\beta}_3 \times \texttt{radio}) \times 1000 = 19 + 1.1 \times \texttt{radio} \text{ units.}$$

- An increase in radio advertising of \$1,000 will be associated with an increase in sales of

$$(\hat{\beta}_2 + \hat{\beta}_3 \times \texttt{TV}) \times 1000 = 29 + 1.1 \times \texttt{TV} \text{ units.}$$

Hierarchy

- Sometimes it is the case that an interaction term has a very small p-value, but the associated main effects (in this case, TV and radio) do not.
- The hierarchy principle: If we include an interaction in a model, we should also
  include the main effects, even if the p-values associated
  with their coefficients are not significant.
- The rationale for this principle is that interactions are hard to interpret in a model without main effects - their meaning is changed. Specifically, the interaction terms also contain main effects, if the model has no main effect terms.

Interactions between qualitative and quantitative variables

Consider the Credit data set, and suppose that we wish to predict balance using income (quantitative) and student (qualitative).

Without an interaction term, the model takes the form

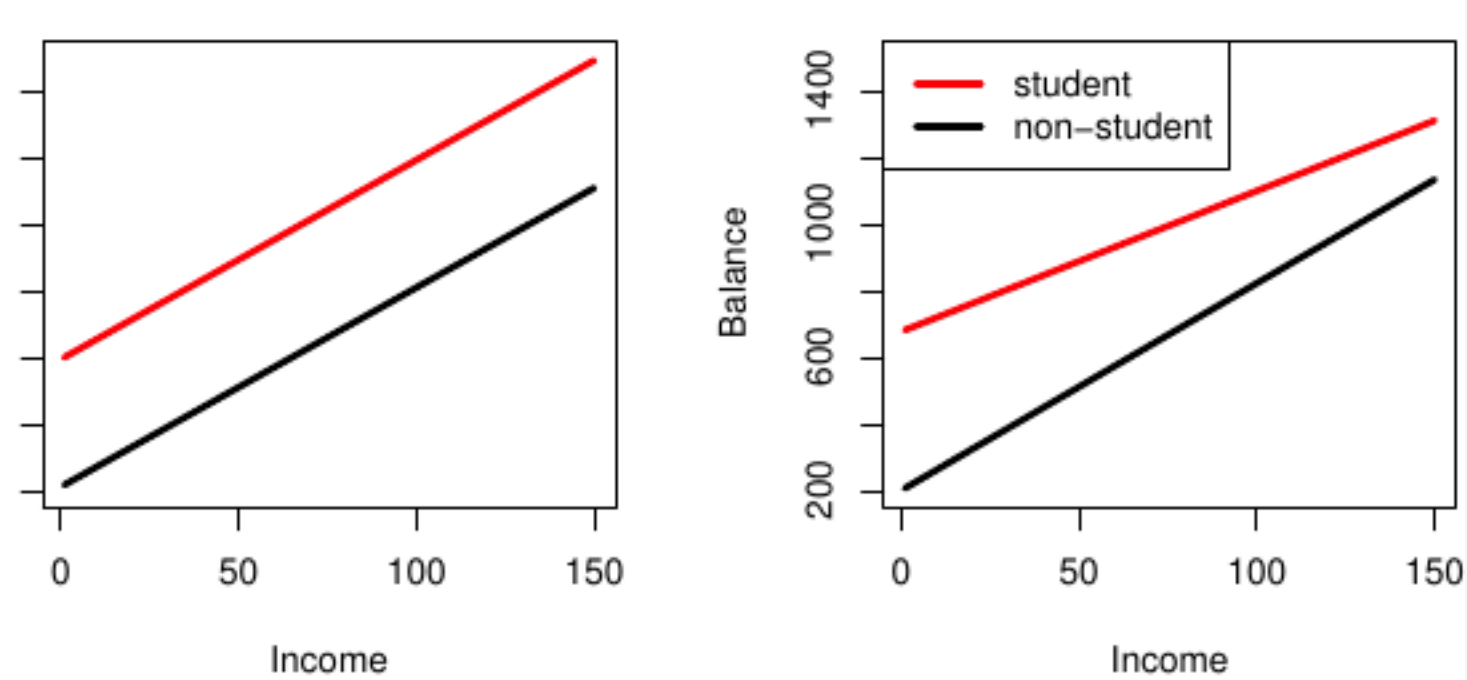
$$\text{balance}_i \approx \beta_0 + \beta_1 \times \text{income}_i + \begin{cases} \beta_2 & \text{if } i\text{th person is a student} \\ 0 & \text{if } i\text{th person is not a student} \end{cases}$$

$$= \beta_1 \times \text{income}_i + \begin{cases} \beta_0 + \beta_2 & \text{if } i\text{th person is a student} \\ \beta_0 & \text{if } i\text{th person is not a student.} \end{cases}$$

With interactions, it takes the form

$$\text{balance}_i \approx \beta_0 + \beta_1 \times \text{income}_i + \begin{cases} \beta_2 + \beta_3 \times \text{income}_i & \text{if student} \\ 0 & \text{if not student} \end{cases}$$

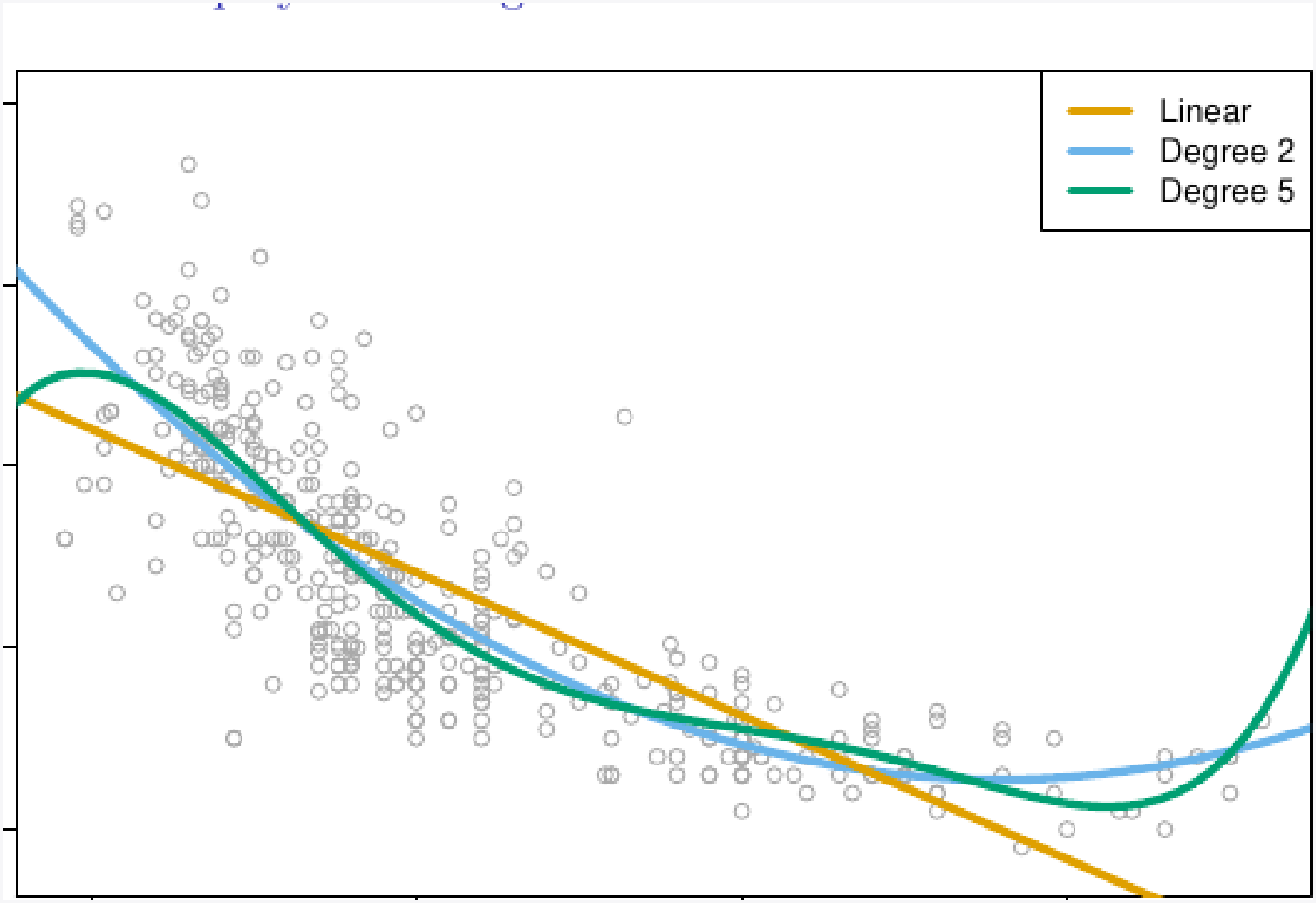
$$= \begin{cases} (\beta_0 + \beta_2) + (\beta_1 + \beta_3) \times \text{income}_i & \text{if student} \\ \beta_0 + \beta_1 \times \text{income}_i & \text{if not student} \end{cases}$$



Credit data; Left: no interaction between `income` and `student`.
Right: with an interaction term between `income` and `student`.

Non-linear effects of predictors

polynomial regression on `Auto` data



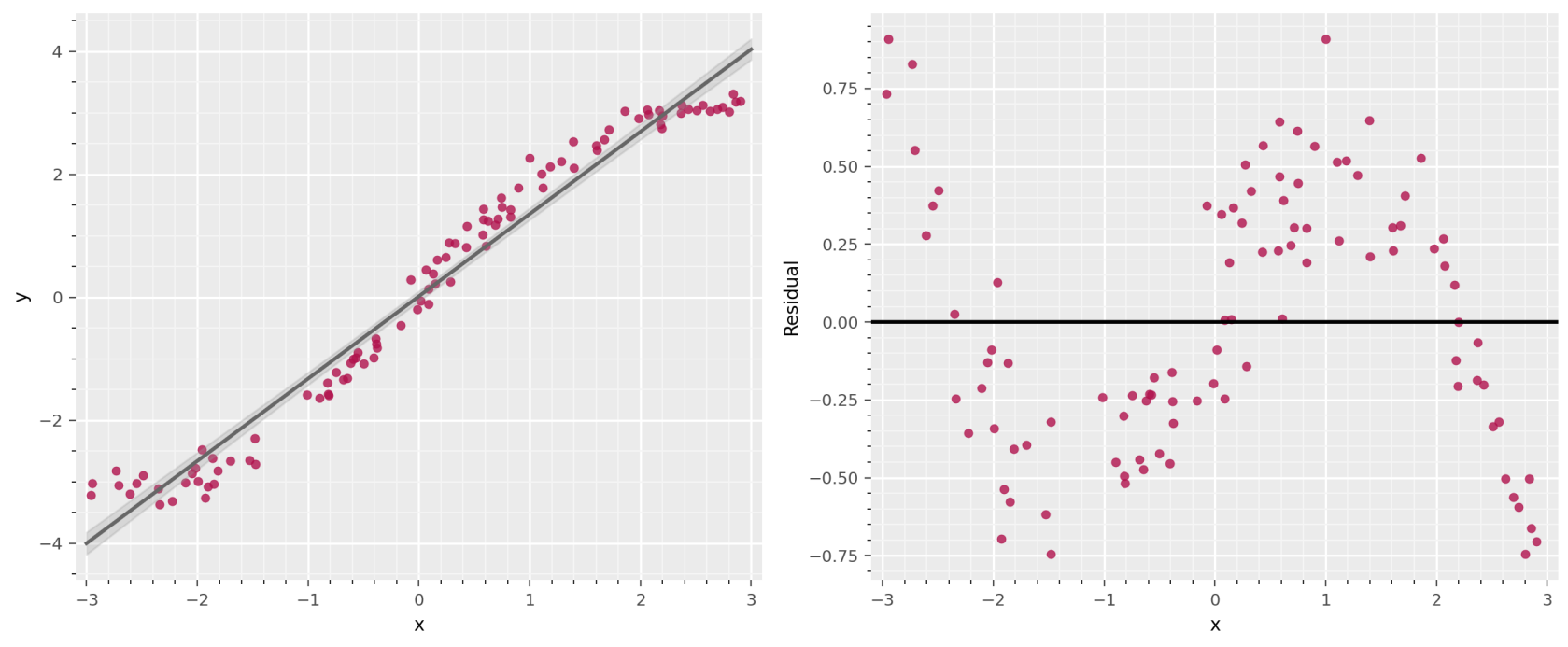
The figure suggests that

$$\text{mpg} = \beta_0 + \beta_1 \times \text{horsepower} + \beta_2 \times \text{horsepower}^2 + \epsilon$$

may provide a better fit.

Coefficient	Std. Error	*t*-statistic	*p*-value	
Intercept	56.9001	1.8004	31.6	< 0.0001
horsepower	-0.4662	0.0311	-15.0	< 0.0001
horsepower^2	0.0012	0.0001	10.1	< 0.0001

Residual plots



Left: simulated data with a regression line. Right: residual plot of the same fitted model.

Interpretation of the residual plots

- The presented data is simulated from

```
x = runif(100, -3, 3); y = x + sin(x) + rnorm(100, sd = .2)
```

- Because $\sin(x)$ is omitted from the fitted linear model, the residuals are not random noise around zero. Instead, they trace the missing nonlinear structure:

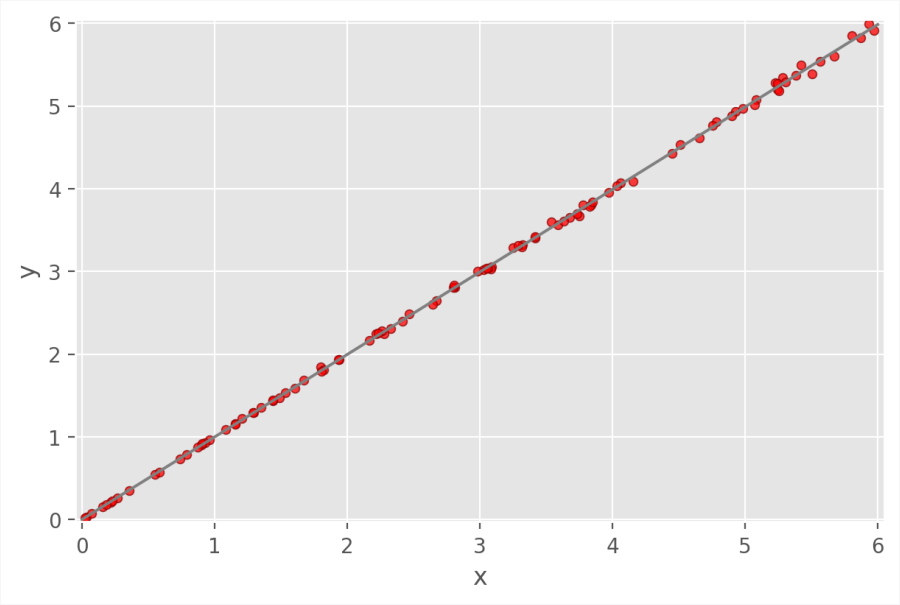
$$e_i \approx \sin(x_i) + \text{noise}.$$

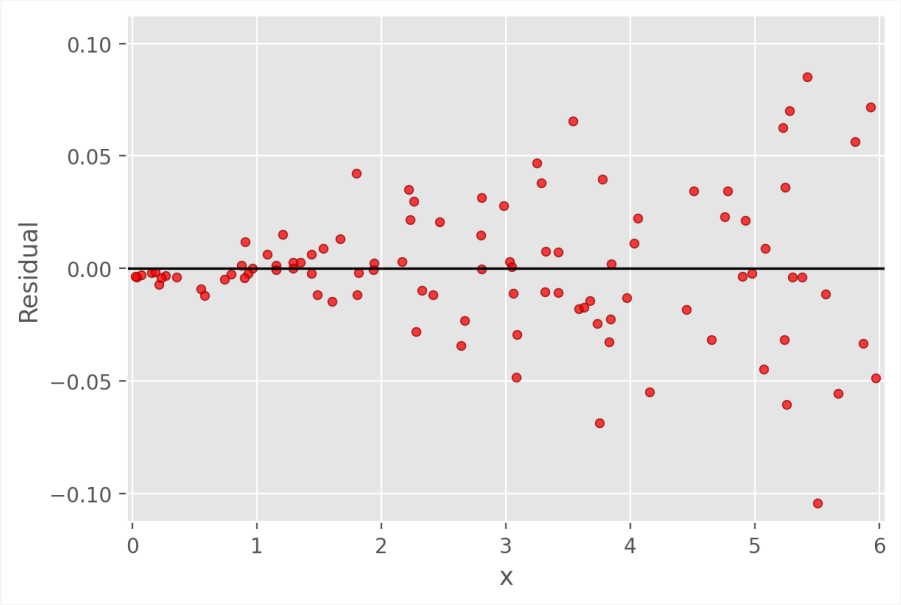
- This is exactly why residual plots are useful: they reveal model misspecification that can be hard to see from the fitted line alone.

Heteroscedasticity

Data are simulated as follows

```
x <- runif(100, 0, 6);  y <- x + rnorm(100, mean = 0, sd = 0.01*x)
```





Left: simulated data with a regression line. Right: residual plot of the same fitted model - variance depends on x.

What should we look for in a residual plot?

- In a well-specified model, residuals should be scattered randomly around 0, with no visible trend or systematic pattern. Any visible pattern is a warning that the model may be inadequate.
- The spread of the residuals should be roughly constant across the range of fitted values or predictors.
- A curved pattern suggests that the mean structure is misspecified, for example because a nonlinear effect is missing.
- A funnel shape suggests heteroscedasticity, that is, non-constant variance.
- Large isolated residuals may indicate outliers or influential observations.
- Distinct groups, bands, or other structure may indicate omitted variables, interactions, or dependence in the data.
- Residual plots do not prove that a model is correct, but they are one of the most important diagnostic tools for detecting when a model is clearly wrong.

March 30, 2026

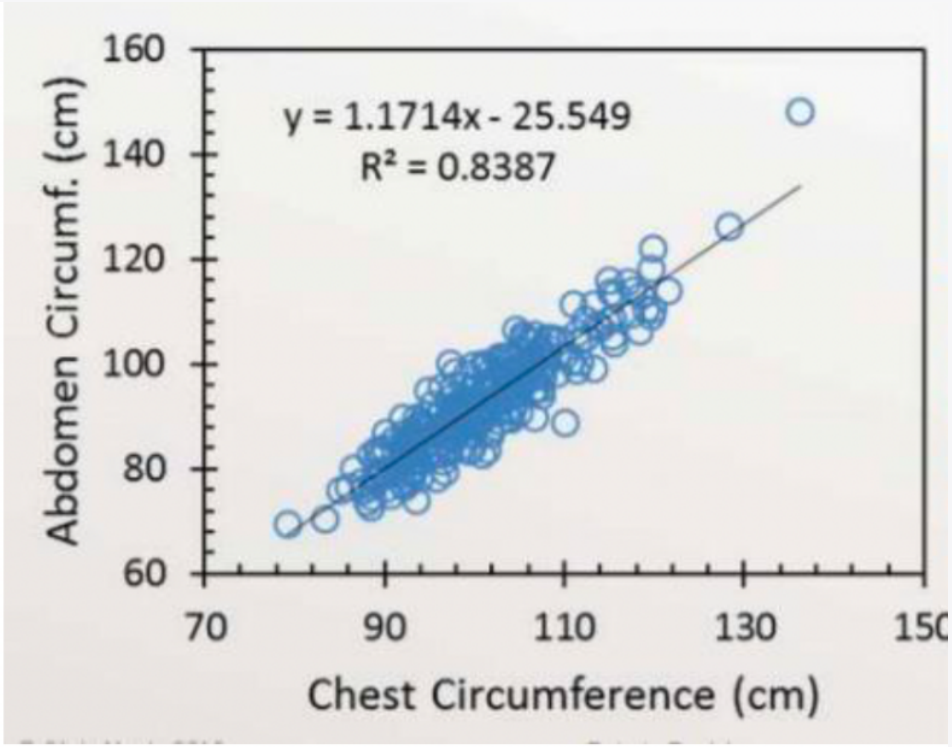
What is multicollinearity?

- Multicollinearity occurs when two or more predictors carry overlapping information and are strongly related to each other. These predictors move together, so their separate effects become hard to disentangle.
- In practice, this often happens because measurements describe similar physical traits, or because the sampled data occupy only a narrow region of the predictor space.

Example. Body fat is hard to measure directly. It is natural to build a model using easier predictors such as weight, height, chest circumference, abdomen circumference, and thigh circumference.

This motivates checking the relationship between **chest circumference** and **abdomen circumference**.

A multicollinearity example



Chest circumference vs. abdomen circumference

- The predictors are strongly related: the scatterplot is nearly linear.

  $\hat{y} = 1.1714x - 25.549, \quad R^2 = 0.8387$

- Strong correlation with the response can be useful, but strong correlation *between predictors* creates instability in coefficient estimates.

	Y	X_1	X_2	X_3
Y	1.00	.945	.836	.701
X_1		1.00	.781	.499
X_2			1.00	.632
X_3				1.00

Correlation coefficients: what they do and do not tell us

- Perfectly correlated predictors should be removed: if one column is an exact linear combination of others, OLS cannot separate their effects.
- Pairwise correlations close to ± 1 are a warning sign for instability in the OLS coefficients.
- In an ideal design, predictors are nearly orthogonal, but this rarely happens in real data.
- Pairwise correlations are only a first check: they can miss more complicated dependence structures.

Important note. The relationship between X and X^2 depends on the range of X. Centering can reduce their correlation, but standardization by itself does *not* solve multicollinearity.

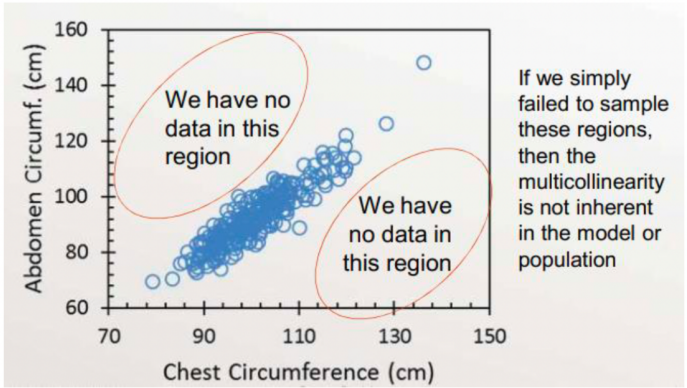
Impact of correlated predictors

- **Prediction can still be good.** If future data have the same correlation structure, fitted values can have low variance and prediction accuracy may remain strong.
- **Individual coefficients become unstable.** Small changes in the data can produce large changes in estimated coefficients, sometimes even reversing signs.
- **Standard errors become large.** Confidence intervals widen and hypothesis tests lose power.
- **Interpretation becomes difficult.** With strongly correlated predictors, the idea of "holding other variables fixed" may no longer be realistic.

Under the usual linear-model assumptions, multicollinearity does *not* by itself make OLS biased; the main problem is high variance and unstable interpretation.

Causes of multicollinearity

- **Sampling:** we only observed a region where the predictors happen to move together.
- **Model or population structure:** some predictors are naturally linked by biology, physics, economics, or accounting identities.
- **Model choice:** a different model specification might describe the data better.
- The issue often becomes more visible when the predictor range is small; for example, X and X^2 can look highly correlated over a narrow interval.



Missing regions can create an apparent collinearity problem

How to detect multicollinearity

- **Correlation matrix:** quick and easy, but only captures simple pairwise relationships.
- **Variance inflation factors (VIF):** measure how much the variance of a coefficient is inflated because the corresponding predictor is explained by the others.
- **Eigensystem analysis:** useful for detecting more complex dependence patterns and motivates principal component analysis (PCA).

A predictor may have only moderate pairwise correlations with every other variable and still be close to a linear combination of several predictors. This is why VIF and eigensystem diagnostics are valuable.

Variance inflation factor (VIF)

$$\mathrm{VIF}_k = \frac{1}{1 - R_k^2}$$

- R_k^2 is the coefficient of determination, obtained by regressing X_k (the k-th predictor) against all of the other predictors.
- If $R_k^2 \to 1$, then most of the information in X_k is already contained in the other predictors, so the variance VIF_k of $\hat{\beta}_k$ blows up to infinity and thus this predictor is redundant.
- If $R_k^2 = 0$, then $\mathrm{VIF}_k = 1$, there is no correlation.

Interpretation. $\mathrm{VIF}_k = 5 \Rightarrow R_K^2 = 0.8$ means the variance of $\hat{\beta}_k$ is five times as large as it would be with uncorrelated predictors. 80% of the variation in X_k can be explained by the other predictors. That is why the coefficient of X_k becomes hard to estimate precisely.
Rule of thumb. Values above 4 merit investigation; values above 10 usually indicate a serious problem, especially when interpretation is the goal.

What to do about multicollinearity?

- If the goal is mainly **prediction**, restrict use of the model to settings where the same correlation pattern is expected.
- **Drop or combine** some correlated predictors, especially variables with the highest VIF values.
- **Collect additional data** that break the collinearity pattern by covering new parts of the predictor space.
- **Use subject-matter knowledge** to measure or fix some coefficients in a separate experiment.
- **Use regularization**, such as Ridge regression.
- **Use dimension reduction**, such as principal component analysis (PCA).

Linear constraints and PCA

- Multicollinearity often comes from constraints among predictors, for example

$$X_2 - X_4 + X_5 = 5, \qquad X_6 = 0.8X_7.$$

- Theoretical considerations can help identify such constraints and simplify the model.
- PCA can help detect unknown linear constraints, but not every dependence is linear; for example,

$$X_2 X_4 = 5$$

  is nonlinear.

Takeaway. Multicollinearity is mainly a problem for *interpreting* coefficients. For pure prediction it can be manageable, but for explanation we need diagnostics, careful design, and sometimes a different model.

Why do we use Ridge and Lasso regression?

Ordinary least squares minimizes $\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$, but when predictors are many or strongly correlated, the fitted coefficients can become unstable and the model may overfit.

- **Ridge regression** minimizes

$$\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^{p} \beta_j^2.$$

- **Lasso regression** minimizes

$$\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^{p} |\beta_j|.$$

Both methods add a penalty to the least-squares criterion, shrink the coefficients toward zero, reduce variance, and often improve prediction on new data.**They are mainly used to address:** overfitting, multicollinearity, unstable or very large coefficient estimates.

Ridge vs. Lasso: main difference and when to use which

- **Ridge regression:**
  - shrinks coefficients toward zero,
  - usually keeps *all* predictors in the model,
  - is especially useful when predictors are strongly correlated.
- **Lasso regression:**
  - shrinks coefficients toward zero,
  - can make some coefficients *exactly zero*,
  - therefore also performs variable selection.

When to use which?

- Use **Ridge** when many predictors may all carry signal and the goal is stable prediction.
- Use **Lasso** when you want a simpler, more interpretable model and believe only some predictors are truly important.

Confounding variable

Correlation between two predictors does *not* automatically mean that a confounding variable exists. Correlated predictors may simply carry overlapping information, which leads to multicollinearity and unstable coefficient estimates.

A **confounding variable** is more specific: it is a variable that is related both to the predictor of interest and to the response, and because of that it can distort the estimated relationship between them. If such a variable is omitted, coefficient interpretation may become biased.

In practice, a plausible confounder should usually be included when the goal is explanation or causal interpretation. For prediction, a variable should be included if it improves predictive performance and is available at prediction time.

Good practice is therefore to distinguish carefully between *correlation among predictors*, *true confounding*, and *other hidden causes*, and not to include variables automatically without thinking about their role in the problem.

Correlated predictors, confounder, and latent variable

Example: studying the effect of study time on exam score

- **Correlated predictors:** suppose we include *study time* and *number of solved practice problems*. These predictors will usually be strongly correlated, because students who study more often also solve more problems. This mainly creates *multicollinearity*: it becomes harder to separate their individual effects, and the coefficient estimates may become unstable.
- **Confounder:** suppose *prior ability* affects both study time and exam score. Stronger students may study differently and also score better. If prior ability is omitted, the estimated effect of study time may be biased, because part of the apparent effect of study time may actually be due to prior ability.
- **Latent variable:** suppose *motivation* affects study time, number of solved problems, and exam score, but motivation is not directly observed. Then motivation is a *latent variable*: an unobserved factor that may induce dependence among the observed variables.

Correlated predictors, confounder, and latent variable

What should we do?

- For **correlated predictors**, check multicollinearity and interpret coefficients carefully.
- For a plausible **confounder**, include it if the goal is explanation or causal interpretation.
- For a **latent variable**, acknowledge the limitation; if possible, measure it indirectly, use proxies, or use a more suitable modeling framework.

Simpson's paradox and why it matters in regression

Simpson's paradox occurs when a trend seen in the aggregated data reverses or changes substantially after the data are split into meaningful groups.

- In the full dataset, predictor X may appear to have a positive association with response Y.
- But after adjusting for a third variable Z (for example, group, department, age, or baseline severity), the association may become weaker, disappear, or even reverse.

Why is this important in regression?

- Regression coefficients are interpreted *conditionally on the variables included in the model*.
- If an important grouping or confounding variable is omitted, the estimated coefficient may reflect a misleading aggregated pattern.
- Thus, regression can give very different conclusions depending on whether relevant variables are included or ignored.

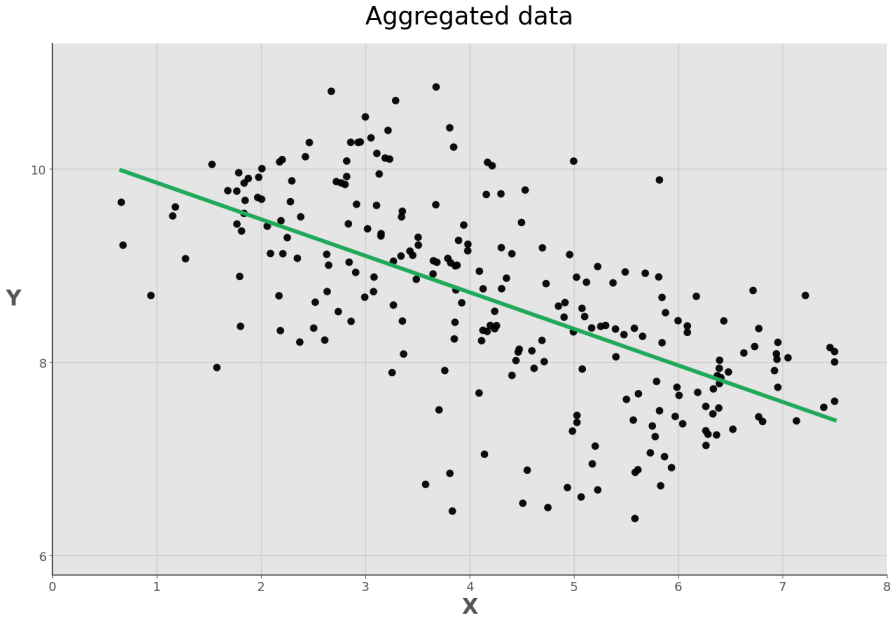
Main lesson: A relationship seen in the overall data is not always the relationship that holds within comparable groups. This is why careful variable selection and adjusted interpretation are essential in regression.

Simpson's paradox

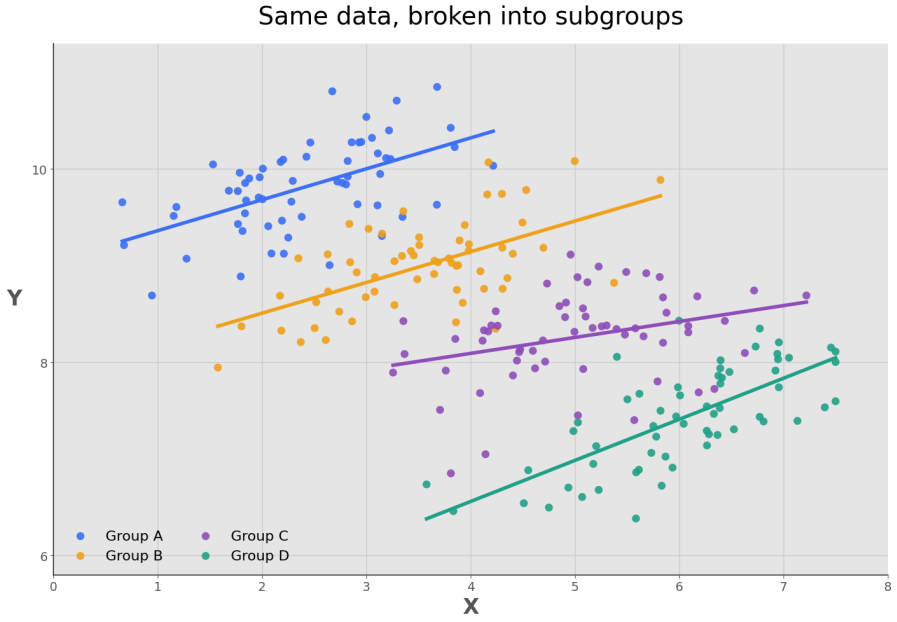
- **Simpson's paradox** occurs when the overall association in aggregated data has a different direction from the association seen within relevant subgroups.
- The paradox is usually created by a **third variable** that acts as a confounder.
- If we aggregate across that confounder, the combined analysis can be misleading.

Key idea: Never decide from the aggregate alone. Always ask whether a subgroup variable changes the story. If we ignore subgroup structure ten we may get a misleading overall trend.

Visual example: aggregate fit vs. subgroup fits



Aggregated data: the overall regression line slopes downward.



After splitting into subgroups: each subgroup has a positive slope.

- The same data can suggest opposite conclusions depending on whether we analyze all observations together or separate them by subgroup.
- The subgroup affects the average value of Y, and it is also associated with different values of X. Because of this, combining the groups can create a misleading overall trend. Therefore the right comparison is *within groups*, not only in the aggregate.

Admissions example: aggregate rates vs. department rates

Example: Admissions rates for men and women at UC Berkeley.

Overall acceptance rates

Gender	Acceptance rate
Men	45%
Women	30%

At first glance, the aggregate suggests that men were admitted at a much higher rate. The percentages are calculated per gender. For ex. 45% means 45% of all men who applied, were admitted.

Largest six departments (rates)

Dept.	Men	Women
A	62%	82%
B	63%	68%
C	37%	34%
D	33%	35%
E	28%	24%
F	6%	7%

Women tended to apply more often to departments with lower overall admission rates, while men applied more often departments with easier enrollment. Department is the confounder, so the meaningful comparison is within department.

Why Simpson's paradox matters?

What the paradox teaches us? Aggregation can hide subgroup structure. A summary regression line or summary rate can reverse the story. The paradox is a warning about confounding, omitted variables, and interpretation.

What to be careful about? Do not interpret the aggregate automatically. Check whether the groups differ both in outcome level and in the predictor distribution. For explanation, ask which variable may be acting as the confounder.

How to deal with it? Plot the aggregate and the subgroup analyses side by side. Stratify the analysis by the suspected confounder. In regression, include the confounder explicitly. Report both overall and adjusted results when helpful.

Bottom line: Simpson's paradox shows that aggregation without context can mislead, especially when the subgroup variable affects both the predictor and the outcome.

Important topics not discussed in detail

Regression assumptions. Example: Suppose we model apartment price from size. If the true relationship bends upward for large apartments, a straight-line model may systematically underpredict expensive apartments. If observations come from repeated measurements on the same building, the errors may also fail to be independent.

What to do: Check whether the linear form is reasonable by using scatterplots and residual plots. Check independence based on how the data were collected; for time-ordered or repeated observations, independence may fail. Check whether the variance of the residuals is roughly constant across the fitted values. For inference, examine whether the residuals are approximately normal, for example by a QQ-plot. Normality is mainly important for tests and confidence intervals, not for fitting the least-squares line itself.

Important topics not discussed in detail

Outliers, leverage, and influential observations. Example: In a dataset of salaries versus years of experience, one executive with an extremely high salary may be an outlier. A person with 40 years of experience, far outside the range of the other data, is a high-leverage point. Such observations may strongly change the fitted line.

What to do: Check residuals to identify outliers, leverage values to identify observations with unusual predictor values, and measures such as Cook's distance to detect influential points. Investigate such observations carefully: they may be recording errors, unusual but valid cases, or indications that the model form is not appropriate. They should not be removed automatically; first understand why they are unusual and how strongly they affect the fitted model.

Important topics not discussed in detail

Additional diagnostic plots. Example: A QQ-plot can reveal that residuals have heavy tails, so t-tests and confidence intervals may be less reliable. A leverage or Cook's distance plot can show that one or two points dominate the fitted model, even if the residual plot looks acceptable.

What to do: In addition to the residual-versus-fitted plot, inspect QQ-plots for normality, scale-location plots for changing variance, and leverage/Cook's distance plots for influential cases. No single plot is sufficient on its own; diagnostics should be interpreted together.

Why this matters?: A regression line may look reasonable, but hidden violations of assumptions or a few unusual points can make interpretation and inference misleading.

Important topics not discussed in detail

Validation on unseen data. Example: A model predicting exam scores from many student characteristics may fit the training data very well, but perform poorly on new students. Good in-sample fit does not guarantee good generalization.

What to do: Evaluate predictive performance on data not used for fitting, for example by using a test set or cross-validation. Compare models not only by training R^2 or residual error, but also by how well they predict new observations. This is especially important when many predictors are included.

Model selection and regularization. Example: In a medical study with many correlated predictors such as blood measures, age, and body measurements, the fitted coefficients may become unstable. Methods such as Ridge and LASSO help control complexity and improve prediction.

What to do: Avoid relying only on stepwise procedures or p-values. Consider adjusted R^2, AIC/BIC, cross-validation, and regularization methods such as Ridge or LASSO. These are particularly useful when predictors are numerous, highly correlated, or when prediction is more important than interpretation of individual coefficients.

Important topics not discussed in detail

Causal interpretation. Example: A regression may show that students who spend more hours studying obtain higher grades, but this does not by itself prove that increasing study time causes the improvement. Motivation, prior knowledge, or family support may also affect both variables.

What to do: Interpret regression coefficients primarily as associations unless the study design supports a causal claim. Think about possible confounders, omitted variables, selection effects, and the direction of the relationship. If the goal is causal interpretation, stronger assumptions or a more suitable study design are needed.

Why this matters?: In practice, we want models that are not only mathematically fitted, but also stable, predictive on new data, and interpreted with appropriate caution.